



UNIVERSIDAD NACIONAL
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Project: Lexer (1st Submission)

Course: Compilers

Group 5

Team 04

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Public Repository:

<https://github.com/HansMartza/unam.fi.compilers.g5.04.git>

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1 Introduction

This project implements a Lexical Analyzer (Lexer) developed in Python. It was built using the programming language's native regular expressions library (`re`).

The main package and module was named `unam.fi.compilers.g5.04`. The `Lexer` class was designed to receive and inspect the input data by reading a text string.

2 Lexer Specifications: Categories and Mapping

The lexical analyzer processes more than 40 tokens during the execution tests and complies with the inclusion of the mandatory basic token `print`.

Mapping / Explanatory Legend

The lexer breaks down and classifies the found tokens by assigning them one of the following tags:

- **keyword:** Corresponds to the language's reserved words (including `print`, `int`, `float`, `string`, etc.).
- **identifier:** Corresponds to identifiers, such as variable names.
- **operator:** Corresponds to mathematical and assignment operators (`+`, `-`, `=`, etc.).
- **constant:** Corresponds to constants and numerical values.
- **punctuation:** Corresponds to punctuation marks and delimiters (`;`, `()`, `{}`, etc.).
- **literal:** Corresponds to text strings enclosed in quotes.

3 Theoretical Development and Grammar

Code Example

The following code block was used as the basis for the theoretical demonstration:

```
1 print("This is an example!!!!");
2 int b = 20;
3 float c = (500 * 4) / 20 + 1 - 10;
4 print(c);
5 print("hello");
6 string answer_5 = c;
```

Grammar with Left Factoring

$$\begin{aligned} \langle \text{program} \rangle &\rightarrow \langle \text{statement_list} \rangle \\ \langle \text{statement_list} \rangle &\rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle \mid \varepsilon \\ \langle \text{statement} \rangle &\rightarrow \langle \text{declaration} \rangle ; \mid \langle \text{print_statement} \rangle ; \\ \langle \text{declaration} \rangle &\rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle \\ \langle \text{type} \rangle &\rightarrow \text{int} \mid \text{float} \mid \text{string} \\ \langle \text{print_statement} \rangle &\rightarrow \text{print} (\langle \text{value} \rangle) \\ \langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \mid \langle \text{literal} \rangle \\ \langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\ \langle \text{expression}' \rangle &\rightarrow + \langle \text{term} \rangle \langle \text{expression}' \rangle \mid - \langle \text{term} \rangle \langle \text{expression}' \rangle \mid \varepsilon \\ \langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\ \langle \text{term}' \rangle &\rightarrow * \langle \text{factor} \rangle \langle \text{term}' \rangle \mid / \langle \text{factor} \rangle \langle \text{term}' \rangle \mid \varepsilon \\ \langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \mid \langle \text{identifier} \rangle \mid (\langle \text{expression} \rangle) \\ \langle \text{literal} \rangle &\rightarrow " \langle \text{content} \rangle " \\ \langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \mid \varepsilon \\ \langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\ \langle \text{identifier}' \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \mid \langle \text{digit} \rangle \langle \text{identifier}' \rangle \mid \varepsilon \\ \langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\ \langle \text{constant}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \mid \varepsilon \\ \langle \text{letter} \rangle &\rightarrow a \mid \dots \mid z \mid A \mid \dots \mid Z \mid _ \\ \langle \text{digit} \rangle &\rightarrow 0 \mid 1 \mid \dots \mid 9 \\ \langle \text{character} \rangle &\rightarrow \text{any symbol except } " \end{aligned}$$

4 Theoretical Verification of Derivations

Instruction: `print("This is an example!!!!");`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{print_statement} \rangle ;$
 $\langle \text{print_statement} \rangle \rightarrow \text{print} (\langle \text{value} \rangle)$
 $\langle \text{value} \rangle \rightarrow \langle \text{literal} \rangle$
 $\langle \text{literal} \rangle \rightarrow " \langle \text{content} \rangle "$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{T}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{h}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{i}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{s}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow _$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{i}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{s}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow _$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{a}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{n}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow _$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{e}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{x}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{a}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{m}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$

$$\begin{aligned}
\langle \text{character} \rangle &\rightarrow p \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow l \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow e \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \langle \text{character} \rangle \langle \text{content} \rangle \\
\langle \text{character} \rangle &\rightarrow ! \\
\langle \text{content} \rangle &\rightarrow \varepsilon \\
\langle \text{statement_list} \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `int b = 20;`

$$\begin{aligned}
\langle \text{program} \rangle &\rightarrow \langle \text{statement_list} \rangle \\
\langle \text{statement_list} \rangle &\rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle \\
\langle \text{statement} \rangle &\rightarrow \langle \text{declaration} \rangle ; \\
\langle \text{declaration} \rangle &\rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle \\
\langle \text{type} \rangle &\rightarrow \text{int} \\
\langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\
\langle \text{letter} \rangle &\rightarrow b \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \\
\langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \\
\langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 2 \\
\langle \text{constant}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 0 \\
\langle \text{constant}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `float c = (500 * 4) / 20 + 1 - 10;`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{declaration} \rangle ;$
 $\langle \text{declaration} \rangle \rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle$
 $\langle \text{type} \rangle \rightarrow \text{float}$
 $\langle \text{identifier} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{c}$
 $\langle \text{identifier}' \rangle \rightarrow \varepsilon$
 $\langle \text{value} \rangle \rightarrow \langle \text{expression} \rangle$
 $\langle \text{expression} \rangle \rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle$
 $\langle \text{term} \rangle \rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow (\langle \text{expression} \rangle)$
 $\langle \text{expression} \rangle \rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle$
 $\langle \text{term} \rangle \rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow \langle \text{constant} \rangle$
 $\langle \text{constant} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 5$
 $\langle \text{constant}' \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 0$
 $\langle \text{constant}' \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 0$
 $\langle \text{constant}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow * \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow \langle \text{constant} \rangle$
 $\langle \text{constant} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 4$
 $\langle \text{constant}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow \varepsilon$
 $\langle \text{expression}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow / \langle \text{factor} \rangle \langle \text{term}' \rangle$
 $\langle \text{factor} \rangle \rightarrow \langle \text{constant} \rangle$
 $\langle \text{constant} \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 2$
 $\langle \text{constant}' \rangle \rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle$
 $\langle \text{digit} \rangle \rightarrow 0$
 $\langle \text{constant}' \rangle \rightarrow \varepsilon$
 $\langle \text{term}' \rangle \rightarrow \varepsilon$
 $\langle \text{expression}' \rangle \rightarrow + \langle \text{term} \rangle \langle \text{expression}' \rangle$

$$\begin{aligned}
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \\
\langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 1 \\
\langle \text{constant}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow -\langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{constant} \rangle \\
\langle \text{constant} \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 1 \\
\langle \text{constant}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{constant}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 0 \\
\langle \text{constant}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `print(c);`

$$\begin{aligned}
\langle \text{program} \rangle &\rightarrow \langle \text{statement_list} \rangle \\
\langle \text{statement_list} \rangle &\rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle \\
\langle \text{statement} \rangle &\rightarrow \langle \text{print_statement} \rangle ; \\
\langle \text{print_statement} \rangle &\rightarrow \text{print} (\langle \text{value} \rangle) \\
\langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \\
\langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{identifier} \rangle \\
\langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\
\langle \text{letter} \rangle &\rightarrow c \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

Instruction: `print("hello");`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{print_statement} \rangle ;$
 $\langle \text{print_statement} \rangle \rightarrow \text{print} (\langle \text{value} \rangle)$
 $\langle \text{value} \rangle \rightarrow \langle \text{literal} \rangle$
 $\langle \text{literal} \rangle \rightarrow " \langle \text{content} \rangle "$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{h}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{e}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{l}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{l}$
 $\langle \text{content} \rangle \rightarrow \langle \text{character} \rangle \langle \text{content} \rangle$
 $\langle \text{character} \rangle \rightarrow \text{o}$
 $\langle \text{content} \rangle \rightarrow \varepsilon$

Instruction: `string answer_5 = c;`

$\langle \text{program} \rangle \rightarrow \langle \text{statement_list} \rangle$
 $\langle \text{statement_list} \rangle \rightarrow \langle \text{statement} \rangle \langle \text{statement_list} \rangle$
 $\langle \text{statement} \rangle \rightarrow \langle \text{declaration} \rangle ;$
 $\langle \text{declaration} \rangle \rightarrow \langle \text{type} \rangle \langle \text{identifier} \rangle = \langle \text{value} \rangle$
 $\langle \text{type} \rangle \rightarrow \text{string}$
 $\langle \text{identifier} \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{a}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{n}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{s}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{w}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{e}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{r}$
 $\langle \text{identifier}' \rangle \rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle$
 $\langle \text{letter} \rangle \rightarrow \text{-}$

$$\begin{aligned}
\langle \text{identifier}' \rangle &\rightarrow \langle \text{digit} \rangle \langle \text{identifier}' \rangle \\
\langle \text{digit} \rangle &\rightarrow 5 \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{value} \rangle &\rightarrow \langle \text{expression} \rangle \\
\langle \text{expression} \rangle &\rightarrow \langle \text{term} \rangle \langle \text{expression}' \rangle \\
\langle \text{term} \rangle &\rightarrow \langle \text{factor} \rangle \langle \text{term}' \rangle \\
\langle \text{factor} \rangle &\rightarrow \langle \text{identifier} \rangle \\
\langle \text{identifier} \rangle &\rightarrow \langle \text{letter} \rangle \langle \text{identifier}' \rangle \\
\langle \text{letter} \rangle &\rightarrow c \\
\langle \text{identifier}' \rangle &\rightarrow \varepsilon \\
\langle \text{term}' \rangle &\rightarrow \varepsilon \\
\langle \text{expression}' \rangle &\rightarrow \varepsilon
\end{aligned}$$

5 Program Execution Results

Below is the output generated by the lexical analyzer upon receiving the input data. This breakdown verifies the expected output format and the total count.

The lexical errors shown at the end of the output (`Character not recognized '!'`) were intentionally placed in the test string using the instruction `print(!!!"hello");`. The purpose of this is to demonstrate the program's behavior when a character that does not match any of the defined tokens is entered, verifying that the `errorHandler` class correctly captures the event and reports the exact line and column of the failure.

Output generated by the program:

```

1 -----Tokens List-----
2 keyword int
3 identifier a
4 operator =
5 constant 10
6 punctuation ;
7 keyword print
8 punctuation (
9 literal "This is an example!!!!"
10 punctuation )
11 punctuation ;
12 keyword int
13 identifier b
14 operator =
15 constant 20
16 punctuation ;
17 keyword float
18 identifier c
19 operator =
20 punctuation (
21 constant 500
22 operator *

```

```
23 constant 4
24 punctuation )
25 operator /
26 constant 20
27 operator +
28 constant 1
29 operator -
30 constant 10
31 punctuation ;
32 keyword print
33 punctuation (
34 identifier c
35 punctuation )
36 punctuation ;
37 keyword print
38 punctuation (
39 literal "hello"
40 punctuation )
41 punctuation ;
42 keyword string
43 identifier answer_5
44 operator =
45 identifier c
46 punctuation ;
47
48 *Total of tokens: 45
49
50 --- Report of errors ---
51 Lexical Error [Line 6, Column 6]: Character not recognized '!'
52 Lexical Error [Line 6, Column 7]: Character not recognized '!'
53 Lexical Error [Line 6, Column 8]: Character not recognized '!'
```